



A 'low-tech' Academic Virtue Ethics in the Age of Generative AI

Timothy Daly¹

Received: 2 July 2025 / Accepted: 31 August 2025 / Published online: 22 October 2025
© The Author(s), under exclusive licence to Springer Nature B.V. 2025

Abstract

An academic must both read and write papers. Generative artificial intelligence (AI) can now do both. I draw on Aristotle's *Nicomachean Ethics* to argue that reliance on AI outsources activities and reduces virtues necessary to achieve the purpose of academia—flourishing through the cultivation of reason. I highlight empirical research suggesting the threat posed by reliance on AI for the cultivation of intellectual virtues, and the need for sustained engagement with intellectual tasks. I thus urge scholars to resist delegating academic tasks to AI and instead to adopt a 'low-tech' virtue ethics, for which I offer advice on practical implementation.

Keywords virtue ethics · low-tech · flourishing · academia · aristotle · artificial intelligence · writing · reading

Introduction: Flourishing as the End Goal of Academic Life

A life in academia is ridden with obstacles and frustration, but also moments of joy. I follow the Ancient Greek philosopher Aristotle in claiming that the ultimate goal or 'telos' of academic life is flourishing or 'eudaimonia' through the cultivation of the faculty of reason. Five intellectual virtues are the means by which flourishing is achieved through habit, required for excellence in all of the virtues. I describe these virtues before outlining the threats posed to them by generative artificial intelligence (AI) in the form of large language models (LLMs) and suggesting the need for a 'low-tech' virtue ethics to promote flourishing in academia in the age of AI. My primary goal is to convince scholars to consider adopting this ethic so as to cultivate their own intellectual virtues. My secondary goal, which is much broader and beyond the scope of this opinion paper, is to encourage reflection around why and how other actors should promote the intellectual virtues in academia, e.g. for educators to cultivate virtue in their students as their primary function, and for decision makers in academic policy to reward virtue (rather than metrics of competition, such as publishing outputs).

✉ Timothy Daly
tdaly@flacso.org.ar

¹ Bioethics Program, FLACSO Argentina, Buenos Aires 1966, C1050 AAN, Tucumán, Argentina

Five Intellectual Virtues To Achieve Academic Flourishing

Aristotle—born in 384 BCE and who died in 322 BCE—was an Ancient Greek philosopher and polymath who lectured on almost every domain of knowledge in human thought. A student of Plato, he founded the Peripatetic school of philosophy in the Lyceum in Athens, where he began the wider Aristotelian tradition that set the groundwork for the development of modern science.

Aristotle lectured extensively on virtue ethics (Aristotle, 1925). The fundamental claim of virtue ethics is that things and actions have a *telos*—a final end. Here, I claim that the end or goal of academic life is flourishing through the cultivation of reason. Intellectual virtues are the means by which an academic can reach that end.

Aristotle identified three classes of intellectual virtues—theoretical (*episteme*, *nous*, *sophia*), practical (*phronesis*), and productive (*techné*). *Episteme* is knowledge of one's academic field, its aims, methods, and the conclusions drawn from them. *Nous* is intuition based on that knowledge—it is about knowing which ideas are good ideas, and making judgements within the field based on one's expertise. *Sophia* is a combination of *episteme* and *nous*, a higher form of wisdom that is not merely related entirely to one's field, but rather how one's field relates to the rest of life and society. As a practical virtue, *phronesis* is the ability to decide how to reach a certain end, and also the ability of reason to determine what counts as a desirable end. Finally, the productive virtue of *techné* is the ability to make something through the use of true reasoning. Across all domains of academia, scholars tend to be artisans of academic papers, which when published are read, discussed and cited by academic and broader society.

All five of these intellectual virtues must be practiced through habit, and have been throughout history, to achieve the end of flourishing through the cultivation of reason. To this end, students are given tasks that academics perform throughout their academic careers. Relevant tasks include reading the literature for relevant concepts and papers; reading and summarising important and/or recent papers; and producing a paper on a given topic, including the use of statistical analysis in technical domains. All of these tasks cultivate the five intellectual virtues. However, with the recent arrival of LLMs, all five of them are vulnerable to being diminished by the use of generative AI.

The Threat of Generative Artificial Intelligence To Intellectual Virtue in Academia

Professors and students alike can now use generative AI to perform most intellectual tasks that they would previously have undertaken themselves, from literature research, reading, writing, and performing statistical analyses. This means that AI can now perform any or all tasks involved in academic life.

However, LLMs function through “prompts,” a goal-oriented command. This means that they need a human prompter through which to achieve a certain end. Academics with some degree of familiarity with their field and its norms will know which prompts are necessary to achieve a convincing end. Anecdotal and bibliometric evidence suggest that LLMs are being increasingly used by students and academics alike to produce papers, suggesting that an increasing amount of intellectual tasks are being delegated to AI (Joelving, 2024; Kobak et al., 2025).

This increasing reliance on generative AI amounts to a community-wide voluntary replacement of certain intellectual tasks. This has important negative implications for academia,

and also for the individuals involved in using LLMs to perform certain tasks. Fasoli et al. (2025) have recently provided a framework for the study of technology-induced cognitive diminishment (TICD) that applies to both single episodes of cognitive performance and longer-term cognitive abilities. They summarize several empirical and review studies (ibid., Fig. 2) providing evidence that TICD applies to working memory; attention, learning and metacognition; and spatial abilities. Moreover, in a recent preprint, Kosmyna and colleagues propose a mechanism linking short-term TICD to long-term TICD: the empirically-grounded concept of 'cognitive debt' in AI use, suggesting that task delegation in writing essays in an academic context leads to underperformance "at neural, linguistic, and behavioral levels" over a 4-month period (Kosmyna et al., 2025). These correlates provide material evidence of the undermining of the cultivation or maintenance of intellectual virtues by reliance on AI to perform academic tasks (Table 1).

Table 1 Five intellectual virtues for academic flourishing and their vulnerability to reliance on artificial intelligence to perform academic tasks. Author's note: scholars of Aristotle may disagree around the precise meaning of these terms, whose interpretations here are related to the virtues' function in contemporary academia in light of the potential impacts of AI on them

Intellectual virtue described by Aristotle	Description in the context of contemporary academia	Vulnerability to task delegation to artificial intelligence (AI)
Episteme	Knowledge of one's academic field, its aims, methods, and the conclusions drawn from them.	Instead of reading original papers, researchers may instead rely on AI to provide partial summaries.
Nous	Intuition based on episteme—knowing which ideas are good ideas, and making judgements within the field based on one's expertise.	Lack of sustained engagement with original research may diminish the researcher's intuition about the validity of certain ideas and conclusions present in a work.
Sophia	A combination of episteme and nous, a higher form of wisdom that is not merely related entirely to one's field, but rather how one's field relates to the rest of life and society.	Sustained use of AI may diminish the researcher's ability to discern the right relationship between research, life and society.
Phronesis	The ability to decide how to reach a certain end, and also the ability of reason to determine what counts as a desirable end.	Reliance on AI may lead to an increasing reliance on shortcuts to achieve certain ends, and bring a sense of meaninglessness if the virtuous end of the academic life is overlooked.
Techne	The ability to make something through the use of true reasoning. Across all domains of academia, scholars tend to be artisans of academic papers, which when published are read, discussed and cited by academic and broader society.	Empirical evidence suggests that academics are increasingly using AI to write papers, favouring their publication metrics rather than advancing scholarly dialogue.

A 'low-tech' Ethic To Promote Virtue in Academia in the Age of AI

What can be done to promote intellectual virtue when any or all academic tasks can be replaced by AI? I consider that a 'low-tech' philosophy is necessary to restore academic virtue:

““Low-tech” is not a noun but an adjective ... a low-tech mindset is simply one that is critical about new technologies. It refuses to assume that a new technology is inherently better than the one it replaces. A low-tech approach also questions the criteria which make us decide whether or not something is “better” than something else” (De Decker, 2025).

Some degree of technology—whether it be pen-and-paper, a typewriter or a computer—is a necessary part of academic life. A low-tech philosophy encourages researchers to use no more tech than necessary to cultivate academic flourishing. This means not using LLMs for any task delegation. This, however, is a negative framing of what is actually a very positive decision: choosing to use the full spectrum of human potential to perform tasks oriented towards the goal of flourishing. To this end, low-tech is a useful means, and certainly not an end in itself.

Neurocognitive studies suggest the superior brain activation and better learning outcomes associated with using pen and paper instead of digital writing (Ihara et al., 2021; Wiley & Rapp, 2021; Van der Weel & Van der Meer, 2024). Practically speaking, academics could use pen-and-paper to draft their ideas, and typing to write them up into full papers. Instructors and lecturers could encourage and reward sustained engagement with the tasks given to students, such as longhand note-taking in lectures, which has been associated with favourable impacts on attention, memory and emotional regulation (Desselle & Shane, 2018).

However, what about the pragmatic objection that since students are already using AI, should their teachers not simply encourage them to improve the complexity of their prompt writing as something that could contribute to the development of virtuous intellectual habits? I would argue that prompt-writing cannot be a satisfying pedagogical objective of the university because it does not require sustained attention with the entire task of, for example, writing an essay. In this way, reducing a task to prompt-writing is likely to frustrate what Aristotle calls the “proper pleasure” of a task, and thus reduce the likelihood that “Cognitive activities become more discriminating and precise when increased by their proper pleasure” (p. 23, Cagnoli Fieconi, 2020). Increasing the proper pleasure of a task requires sustained attention throughout the entirety of the task. Daniel Kahneman argued in “Thinking, fast and slow” (Kahneman, 2011) that humans have two cognitive systems in place: a fast “System 1” to inform short-term, automatic decision-making and a slow “System 2” for careful thinking about complexity. It is true that honing a prompt would require more sustained engagement and use of intellectual virtues than a simple prompt. But once the student has clicked “Send,” virtue-producing engagement is likely to end there: LLMs produce output at a rate that is likely to distract the prior sustained engagement, and lead to activation of visually-biased, System 1-like automatic processes (Turo et al., 2023). The presence of rapidly-generated, salient content may reinforce other cognitive biases, such as over-rating the LLM output content (Brem & Riveccio, 2024). Thus, prompt-writing is a limited form of training that interrupts sustained engagement with the entirety of a task, and while trained students might become better prompt writers, they are unlikely to become better judges of content. In response to such pitfalls, what is sought is not a ‘low-tech’ ethic for its own sake, but rather, one that increases the likelihood of sustained engagement throughout strenuous cognitive tasks, and reduction of distractions, for the sake of gaining intellectual virtues.

Conclusion

The increasingly widespread use of LLMs in academic publishing is a symptom of perverse incentives that confuse the means of academia (e.g. the publishing of papers) with its ends (i.e. human flourishing through the use of reason). I urge scholars to adopt a 'low-tech' virtue ethics to help them to cultivate their own reasoning abilities. Moreover, I have not addressed the ethics of declaration of intellectual contributions in the age of AI, a separate and pressing ethical dilemma (Kendall, 2024). A low-tech rethink of academia should invite us to reflect on both the means and the ends of individual and collective intellectual endeavours, and invites a more human-centred approach in an increasingly technocratic world.

Authors' Contributions Author 1 contributed to all aspects of the study: conceptualisation, design, writing, validation.

Funding This manuscript received no specific funding.

Data Availability (data transparency)
Not applicable.

Code Availability (software application or custom code)
Not applicable.

Declarations

Ethics Approval Not applicable.

Consent for Publication Not applicable.

Consent To Participate Not applicable.

Conflicts of interest No conflicts of interest to declare.

References

- Aristotle (1925). *Nicomachean ethics* (W.D. Ross, Trans.). The Internet Classics Archive. <https://classics.mit.edu/Aristotle/nicomachaen.html>
- Brem, A., & Riveccio, G. (2024). Artificial intelligence and cognitive biases: A viewpoint. *Journal of Innovation Economics & Management*, 44(2), 223–231. <https://doi.org/10.3917/jie.044.0223>
- Cagnoli Fieconi, E. (2020). Aristotle on attention. *Archiv für Geschichte Der Philosophie*, 103(4), 433–456. <https://doi.org/10.1515/agph-2018-0014>
- De Decker, K. (2025). What is Low-tech Magazine? Low-tech Magazine. Available at: <https://solar.lowtechmagazine.com/about/what-is-low-tech-magazine/>
- Desselle, S. P., & Shane, P. A. (2018). Laptop versus longhand note taking in a professional doctorate course: Student Performance, Attitudes, and behaviors. *Innovations in Pharmacy*, 9(3), 1–12. <https://doi.org/10.24926/iip.v9i3.1392>
- Fasoli, M., Cassinadri, G., & Ienca, M. (2025). The dark side of cognitive enhancement: A framework for the technologically induced cognitive diminishment. *Journal of Cognitive Enhancement*. <https://doi.org/10.1007/s41465-025-00331-7>
- Ihara, A. S., Nakajima, K., Kake, A., Ishimaru, K., Osugi, K., & Naruse, Y. (2021). Advantage of handwriting over typing on learning words: Evidence from an N400 event-related potential index. *Frontiers in Human Neuroscience*, 15, 679191. <https://doi.org/10.3389/fnhum.2021.679191>

- Joelving, F. (2024). *AI-generated commentaries flood journals, distort metrics*. *Science*, 386(6728), 1331–1332. <https://doi.org/10.1126/science.adv4101>
- Kahneman, D. (2011). *Thinking, fast and slow*. Farrar, Straus and Giroux.
- Kendall, G. (2024). When using artificial intelligence tools in scientific publications authors should include the prompts and the generated text as part of the submission. *Journal of Academic Ethics*. <https://doi.org/10.1007/s10805-024-09581-0>
- Kobak, D., González-Márquez, R., Horvát, E. Á., & Lause, J. (2025). Delving into LLM-assisted writing in biomedical publications through excess vocabulary. *Science Advances*, 11(27), eadt3813. <https://doi.org/10.1126/sciadv.adt3813>
- Kosmyna, N., Hauptmann, E., Yuan, Y. T., Situ, J., Liao, X. H., Beresnitzky, A. V., Braunstein, I., & Maes, P. (2025). Your brain on chatgpt: Accumulation of cognitive debt when using an AI assistant for essay writing tasks [Preprint]. arXiv. <https://doi.org/10.48550/arXiv.2506.08872>
- Turo, S., Collin, F., & Brouillet, D. (2023). When the sense of fluency triggers an attentional bias. *Quarterly Journal of Experimental Psychology*, 76(2), 350–360. <https://doi.org/10.1177/17470218221086185>
- Van der Weel, F. R. R., & Van der Meer, A. L. H. (2024). Handwriting but not typewriting leads to widespread brain connectivity: A high-density EEG study with implications for the classroom. *Frontiers in Psychology*, 14, 1219945. <https://doi.org/10.3389/fpsyg.2023.1219945>
- Wiley, R. W., & Rapp, B. (2021). The effects of handwriting experience on literacy learning. *Psychological Science*, 32(7), 1086–1103. <https://doi.org/10.1177/0956797621993111>

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.